

1. Explain the difference between IaaS, PaaS, and SaaS.

IaaS, PaaS, and SaaS are the three main types of cloud computing services.

TYPE	MEANING	Example
IaaS	provides basic infrastructure such as virtual servers, storage, and networking. The user is responsible for managing the operating system, applications, and data.	Amazon EC2
PaaS	provides a ready-made platform where developers can build and deploy applications without worrying about managing servers and operating systems.	AWS Elastic Beanstalk
SaaS	provides complete software applications through the internet. The user does not need to manage the servers or application infrastructure.	Gmail, Power BI service

2. Create and configure an IAM user.

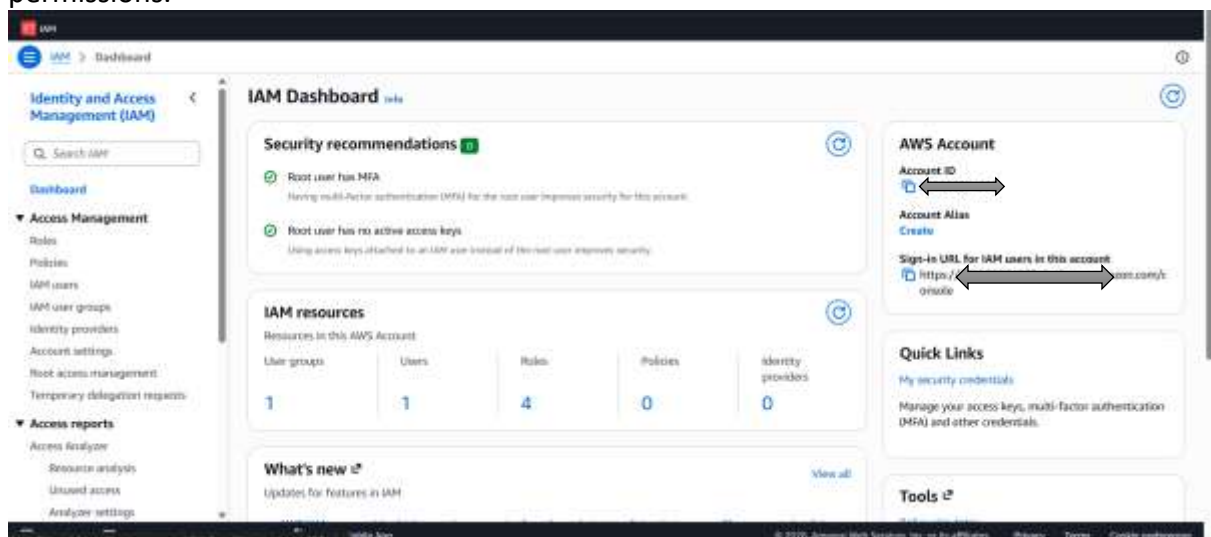
IAM stands for **Identity and Access Management**. It is an AWS service used to control who can access AWS resources and what actions they are allowed to perform.

I created an IAM user by following these steps:

1. I logged in to the **AWS Management Console**.
2. I opened the **IAM service**.
3. I selected **Users** and clicked **Create user**.
4. I entered a username.
5. I configured the required access permissions.
6. I attached an appropriate IAM policy to the user.
7. I reviewed the configuration and created the user.
8. I tested the permissions to make sure the user had only the required access.

For security, I would follow the **principle of least privilege**, which means giving a user only the permissions required to perform their work.

Result: The IAM user was successfully created and configured with the required permissions.



3. Launch an EC2 instance.

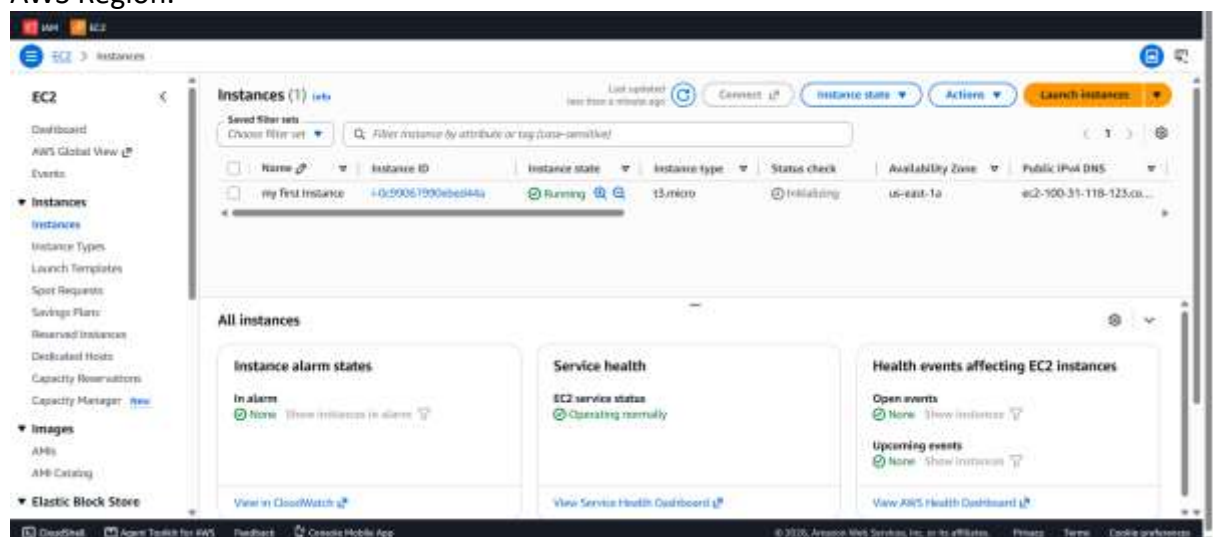
Amazon EC2 stands for **Elastic Compute Cloud**. It provides virtual servers in the AWS Cloud.

I launched an EC2 instance using the following steps:

1. I opened the **AWS Management Console**.
2. I selected the **EC2 service**.
3. I clicked **Launch Instance**.
4. I provided a suitable name for the instance.
5. I selected an appropriate **Amazon Machine Image (AMI)**, such as Amazon Linux.
6. I selected a suitable instance type.
7. I created or selected a **key pair** for secure access.
8. I configured the network settings and security group.
9. I configured the required storage.
10. Finally, I clicked **Launch Instance**.
11. After a few moments, the instance entered the **Running** state.

I then checked the instance details such as its instance ID, public IP address, security group, and storage.

Result: The EC2 instance was successfully launched and was running in the selected AWS Region.



4. Explore AWS regions and availability zones.

AWS has data centre's located in different geographical locations around the world.

These locations are organized into **Regions and Availability Zones (AZs)**.

An **AWS Region** is a geographical area where AWS has multiple data centre's. For example, **Asia Pacific (Mumbai)** is an AWS Region.

An **Availability Zone** is an isolated location within an AWS Region. A Region normally contains multiple Availability Zones.

For example:

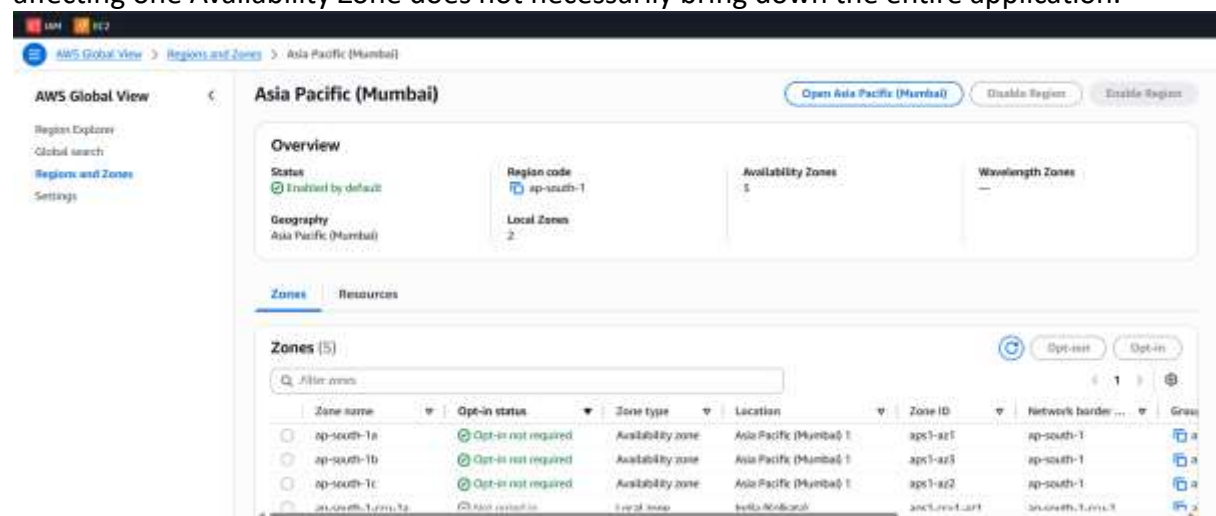
AWS Region → Mumbai

- Availability Zone 1
- Availability Zone 2
- Availability Zone 3

I explored the AWS Management Console and checked the available Regions and their Availability Zones.

Regions and Availability Zones are important because they help with **availability, reliability, disaster recovery, and reducing latency**.

For example, if an application is deployed across multiple Availability Zones, a problem affecting one Availability Zone does not necessarily bring down the entire application.



5. Write a report on key AWS services (EC2, S3, Lambda)

Feature	Amazon EC2	Amazon S3	AWS Lambda
Full Form	Elastic Compute Cloud	Simple Storage Service	AWS Lambda
Main Purpose	Provides virtual servers	Stores files and objects	Runs code without managing servers
Service Type	Compute	Storage	Serverless Compute
What it does	Runs applications on virtual machines	Stores and retrieves data	Executes code when an event occurs
Server Management	I manage the server/OS	No server management	No server management — fully serverless
Example	Hosting a website	Storing images, PDFs, CSV files	Automatically processing an uploaded file
Billing	Mainly based on instance usage	Mainly based on storage and requests	Based mainly on function invocations and execution time
Best suited for	Applications needing full server control	Data storage and backup	Event-driven and automated tasks